

Aliah University

End-Semester Examination (Spring Semester) - 2026

(For 3rd Year 6th Semester BCA Programme)

Paper Name: Machine Learning

Paper Code: CSAUGPC30

Full Marks: 80

Time: 3 hrs

Group – A

5 × 2 = 10

(Answer any five questions)

1. What is the objective of machine learning? [CO1, Remember] 2
2. Define positive and negative correlation. [CO1, Remember] 2
3. What is weather forecasting? [CO1, Understand] 2
4. What is the role of the parameter K in K-Nearest Neighbors (KNN)? [CO2, Understand] 2
5. Define the curse of dimensionality. [CO3, Remember] 2
6. Differentiate between fuzzy clustering and non-fuzzy (hard) clustering. [CO4, Understand] 2

Group - B

(Answer any six questions)

6 × 5 = 30

7. Discuss the applications of machine learning in various real-world domains with suitable examples. [CO1, Understand] 5
8. Differentiate between supervised learning and unsupervised learning with examples. [CO2, Understand] 5
9. Describe the distance calculation methods used in K-Nearest Neighbors (KNN) with examples. [CO2, Apply] 5
10. State and explain Bayes' Theorem used in Naive Bayes classifier. [CO2, Understand] 5
11. Why is feature selection used in machine learning? [CO3, Understand] 5
12. Explain the concept of Principal Component Analysis (PCA). [CO3, Understand] 5
13. Why is a confusion matrix used in machine learning? [CO4, Understand] 5
14. Write a short note on hierarchical clustering. [CO4, Understand] 5

Group – C

(Answer any four questions)

4 × 10 = 40

(All parts of the same question should be answered together)

15. (a) Table 1 shows the distribution of marks obtained by 180 students in an entrance examination. Calculate the mean marks of the students. [CO1, Apply] 6

Mark Range	11 – 20	21 – 30	31 – 40	41 – 50	51 – 60	61 – 70	71 – 80
Frequency	12	28	30	36	34	22	18

Table 1: Marks scored in an entrance examination

- (b) Draw a histogram illustrating the above frequency distribution as shown in Table 1. [CO1, Apply] 4

[Please Turn Over]

16. (a) What is the objective of linear regression? [CO2, Understand] 3
 (b) Using Linear Regression, fit a simple linear regression model for the data given in Table 2 and estimate the sales when advertising cost is ₹6 thousands. [CO4, Apply] 7

X (Advertising Cost in ₹ thousands)	1	2	3	4	5
Y (Sales in ₹ thousands)	10	12	13	15	16

$$m=1$$

$$b=10.2$$

Table 2: Advertising cost vs sales

17. (a) Explain the characteristics of filter methods in feature selection. [CO3, Understand] 5
 (b) Discuss the limitations of filter methods. [CO3, Analyze] 5

18. Consider the data set for a binary classification problem as shown in Table 3:

Sample	Brightness	Saturation	Class
1	40	20	Red
2	50	50	Blue
3	60	90	Blue
4	10	25	Red
5	70	70	Blue
6	60	10	Red
7	25	80	Blue

red.

Table 3: Dataset for a binary classification problem

Predict the class label for a test sample (Brightness = 20, Saturation = 35) using the K-Nearest Neighbors (KNN) classifier. Assume the value of K is 5. [CO4, Apply] 10

19. (a) Explain the concept of a decision tree in detail. [CO4, Understand] 6
 (b) Discuss the advantages of a decision trees. [CO4, Understand] 4
20. (a) Write down the algorithm for K-means clustering. [CO4, Apply] 6
 (b) What are the limitations of K-means clustering? [CO4, Analyze] 4

The curse of Dimensionality is a problem that occurs when the number of features in data becomes very large. As dimensions increase data points become sparse and ML, And it difficult to learn patterns accurately.